COHERENCE IS A TWO-OUT-OF-THREE PROPERTY FOR SHORT EXACT SEQUENCES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author marks the final coherence check 'some details omitted'.

Allowed standard prerequisites (editorial).

- Serre property of coherent modules on locally Noetherian spaces. Cohomology of Spaces, Lemma spaces-cohomology-lemma-coherent-abelian-Noetherian; Sheaves on Stacks, Lemma stacks-sheaves-lemma-quasi-coherent-algebraic-stack. Whole argument.

Source reference. Original reference 06WV: sources/stacks-chow/tags.txt, source lines 8829–8829. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07UC: sources/stacks-chow/tags.txt, source lines 9967–9967. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be a locally Noetherian algebraic stack. Given a short exact sequence $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ in $Mod(\mathcal{O}_\mathcal{X})$ with $\mathcal{F}_1$ and $\mathcal{F}_3$ coherent, then $\mathcal{F}_2$ is coherent.*

Proof. By Sheaves on Stacks, Lemma 06WV part (7) we see that $\mathcal{F}_2$ is quasi-coherent. Then we can check that $\mathcal{F}_2$ is coherent by restricting to $U_{étale}$ for some $U \to \mathcal{X}$ surjective and smooth. This follows from Cohomology of Spaces, Lemma 07UC. Some details omitted. □

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.